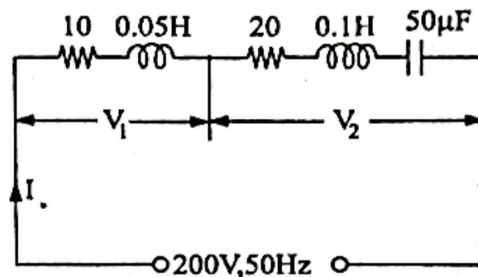


Time: 3 hours

Full Marks: 70

(N. B: Answer any *five* from the following *seven* questions. Figures in the right margin indicate marks allocated for each question)

1. (a) Derive the equation of resonance frequency of series RLC circuit and express the term Quality factor (Q_0). 4
- (b) The bandwidth of a series resonant circuit is 400 Hz. i). If the resonant frequency is 4000 Hz, what is the value of Q_0 ? ii). If $R = 10$, what is the value of X_L at resonance? iii). Find the inductance L and capacitance C of the circuit. 5
- (c) Draw the vector diagram for the circuit shown in Fig., indicating the resistance and reactance drops, the terminal voltages V_1 and V_2 and the current. Find the values of (i) the current I (ii) V_1 and V_2 and (iii) p.f. 5



2. (a) For a source-free series RLC circuit, define the neper frequency (attenuation factor) α and the undamped natural frequency ω_0 . Based on the relationship between α and ω_0 , clearly state the conditions for the circuit to be: 4
 - i. Overdamped
 - ii. Critically damped
 - iii. Underdamped
- (b) Briefly discuss about the unit step, unit impulse and unit ramp functions. 5
- (c) Find $v(t)$ for $t > 0$ in the circuit of Fig.2(c). Assume the switch has been open for a long time and is closed at Calculate at $t=0$, Calculate the $v(t)$ at $t=0.5s$. 5

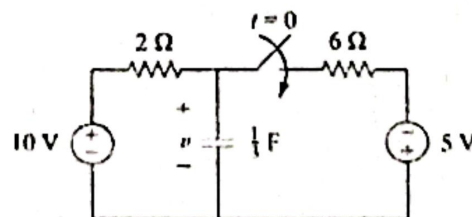


Fig. 2(c) : First order circuit

3. (a) Explain the phase sequence that used in 3 phase system. Show that summation of a balanced three phase circuit voltage is zero. 3
- (b) A balanced abc-sequence Y-connected source with $V_m = 100\angle 10^\circ$ V is connected to a delta connected balanced load $(8+j4) \Omega$ per phase. Calculate the phase and line currents. 5
- (c) For the unbalanced circuit in Fig.3(c), find: (a) the line currents, (b) the total complex power absorbed by the load, 6

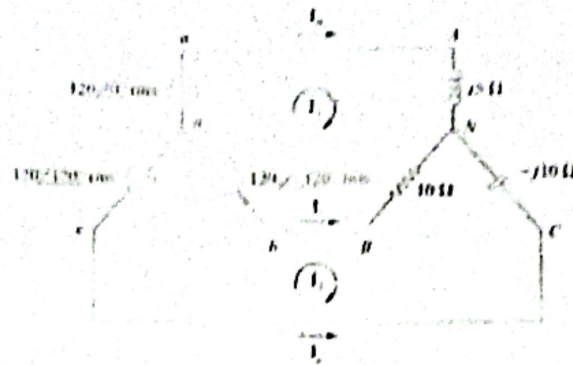


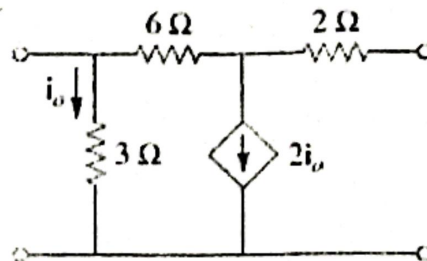
Fig. 3 (c): Unbalanced three phase circuit

4. (a) Define the Coefficient of Coupling (k). Explain the significance of the dot convention in magnetically coupled circuits and state the rule for determining the polarity of the mutually induced voltage. 4
- (b) In the ideal transformer circuit of Fig. 4(b), find V_o and the complex power supplied by the source. 6



Fig. 4(b): Ideal Transformer

- (c) Derive the expression for coupling co-efficient in a magnetically couple circuit. 4
5. (a) Determine the y parameters for the following circuit 4



- (b) Determine the h -parameter of this two port network Fig. 5(b) 6

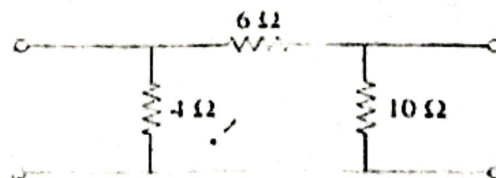


Fig.5(b) Two port Network

- (c) Explain image impedance, iterative impedance and characteristic impedance 4
6. (a) Describe the Two-Wattmeter Method for measuring total power in a three-phase, three-wire system. Prove mathematically that the algebraic sum of the two wattmeter readings ($P_1 + P_2$) exactly equals the total average (real) power absorbed by the load, regardless of whether the load is balanced or unbalanced. 5
- (b) Find the transmission parameter of the two port network Fig. 6(a) 6



Fig. 6(a) Two port Network

- (c) Express h-parameter in terms of z-parameter. 3
7. (a) Draw and design different types of filter circuit and its output. 5
- (b) What type of filter circuit is shown in Fig. 7(b). Calculate the corner or cutoff frequency. 5
- Take $R=2\Omega$, $L=2H$, $C=2\mu F$

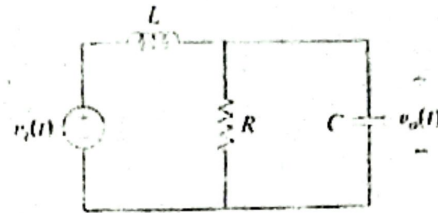


Fig. 7(b): Filter Circuit

- (c) Describe the procedure of designing a constant-k low pass filter with an example. 4